

Syed Ibrahim Mustafa Shah Bukhari

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EDUCATION

Virginia Tech

Expected May 2028

Ph.D. in Computer Science · *Thesis: Privacy in Extended Reality (XR)*

Virginia Tech

August 2023 – May 2026

M.S. in Computer Science · *GPA: 4.00*

Lahore University of Management Sciences (LUMS)

August 2018 – June 2022

B.S. in Computer Science · Minor in Psychology · *Major GPA: 3.92*

RESEARCH EXPERIENCE

Graduate Research Assistant

August 2023 – Present

Virginia Tech

Blacksburg, VA

Advisors: Dr. Brendan David-John and Dr. Bo Ji

PriVisE-Bench: Evaluating Human-Grounded Contextual Privacy Reasoning of Vision-Language Models (VLMs)

- Identified that existing visual-privacy benchmarks model privacy as sensitive-object detection, failing to measure the value alignment of VLMs' privacy judgments with human contextual privacy expectations.
- Reframed scene privacy as a human-grounded contextual judgment and operationalized the evaluation target through privacy label prediction, reasoning alignment, and robustness to prompt wording and ordering.
- Built PriVisE-Bench containing 1,069 whole-scene images spanning eight human-grounded privacy rationales, with image-label associations unanimously validated by three privacy/HCI experts (Gwet's AC1 = 0.914) recruited using purposive sampling.
- Benchmarked 14 VLMs across 326,788 zero-shot, few-shot, and prompt-robustness trials using Holm-corrected statistical tests, finding that only 47.1% of predictions matched both the human label and rationale, exposing alignment failures of current VLMs.
- Impact:** First-author manuscript under review for the *AAAI'27 Special Track on AI Alignment*.

EvaluatAR: Reproducible Cross-Device Evaluation of Bystander Privacy-Enhancing Technologies (PETs) for AR

- Pinpointed that conventional bystander PETs' evaluations rely on live participants, incur substantial setup overhead, and remain device-specific, undermining reproducibility across headsets and slowing iterative development.
- Established a controlled early-stage evaluation protocol that enables researchers and practitioners to reliably assess system feasibility, robustness, and privacy-performance trade-offs under repeatable conditions.
- Engineered EvaluatAR, a modular record-replay framework developed using Unity and Python that standardizes sensor inputs, visual stimuli, runtime measurements, and system outputs across headsets.
- Demonstrated EvaluatAR's cross-device applicability across 2 PETs and 3 headsets, enabling one experimenter to execute 150 controlled trials in about 60 minutes and rapidly prototype and extensively validate fixes for privacy-relevant failures.
- Impact:** First-author paper at *PoPETs'26*; artifact **Available** and **Functional** badges; CCI Research Paper Showcase'26.

Rethinking Privacy Indicators in Extended Reality (XR)

- Recognized that conventional XR privacy indicators assume bystanders are attentive and able to perceive visual cues, leaving distracted or otherwise situationally impaired bystanders unaware of nearby sensing and recording.
- Formulated an inclusive design objective to make XR privacy indicators perceivable to situationally impaired bystanders by adapting their modality, salience, and activation to the surrounding context.
- Led 4 participatory focus groups ($N = 8$), eliciting 7 concepts and consolidating them into 5 novel privacy-indicator designs spanning visual, auditory, haptic, proximity-, gaze-, and activity-triggered cues.
- Assessed the proposed designs through an 8-scenario user study ($N = 7$), finding all to exceed the accepted SUS threshold and multimodal indicators being preferred over visual-only cues in privacy-sensitive impairment scenarios.
- Impact:** First-author paper at *ISMAR-Adjunct'25*.

"Just Stop Doing Everything for Now!": Security Attacks in Collaborative MR

- Exposed a gap in the security of remote collaborative MR, where users' real-time recognition and mitigation of immersive security attacks had not been studied in situ.
- Devised and refined a counterbalanced mixed-methods protocol through 2 pilot rounds, calibrating the collaborative task, attack conditions, behavioral measures, questionnaires, and interview guide.
- Orchestrated the IRB-approved study with 20 participants in 10 remote pairs, capturing task performance, workload, usability, security perceptions, behavioral recordings, and interviews across baseline and attack conditions.
- Triangulated quantitative, observational, and qualitative evidence, finding that click redirection increased average completion time from 5.25 to 8.54 minutes ($p = 0.02$), while 67% of participants exposed to spatial occlusion misreported their actual mitigation.
- Impact:** Published at *IEEE VR'25*; featured by *Computerworld*.

Visceral Interfaces for Privacy Awareness of Eye Tracking in VR

- Investigated whether visceral interfaces could make otherwise invisible VR eye-tracking collection understandable to users without disrupting their experience.
- Designed a counterbalanced mixed-methods study protocol to evaluate 3 Meta Quest Pro interfaces using pre-/post-study privacy measures, scenario surveys, behavioral observations, simulator-sickness screening, and semi-structured interviews.

- Conducted IRB-approved study with 40 participants, enabling comparison of privacy attitudes, usability, and interface preferences.
- Synthesized quantitative and qualitative findings into design recommendations, finding that visceral interfaces significantly increased caution toward sharing gaze data with employers and VR entertainment applications ($p < 0.001$).
- **Impact:** Published at *ISMAR'24*.

Undergraduate Research Assistant
LUMS

January 2022 – May 2023
Lahore, Pakistan

Advisors: Dr. Muhammad Fareed Zaffar and Dr. Murtaza Taj

TaxData: Human-Centered Remote-Sensing Support for Property Tax Surveying

- Identified slow, manual property surveying as a barrier to tax assessment; curated training, validation, and test data from 30,000 Google Earth images, including 3,000 manually labeled constructed and non-constructed examples.
- Developed a remote-sensing pipeline using deep-learning classification and connected-component and YOLO-based plot-segmentation methods, achieving 92% constructed-versus-non-constructed classification accuracy.
- Translated the model into a full-stack decision-support application that enabled tax officials to define regions of interest and inspect current predictions, reducing property-survey time by 70%.
- **Impact:** Featured in *International Growth Centre (IGC March 2022)* technical report.

PUBLICATIONS

- “PriVisE-Bench: Benchmarking Human-Grounded Contextual Privacy Reasoning of Vision Language Models over Whole-Scene Images.” **S. Bukhari**, A. Chinnala, B. Ji, and B. David-John. *AAAI 2027 Special Track on AI Alignment* (under review).
- “EvaluatAR: A Cross-Device Evaluation Framework for Rapid Prototyping of Bystander PETs in AR.” **S. Bukhari**, M. Corbett, B. Ji, and B. David-John. *Proceedings on Privacy Enhancing Technologies (PoPETs 2026(4))*.
- “Rethinking Privacy Indicators in Extended Reality: Multimodal Design for Situationally Impaired Bystanders.” **S. Bukhari**, M. Sajid, B. Ji, and B. David-John. *2025 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct 2025)*.
- “Just Stop Doing Everything for Now!”: Understanding Security Attacks in Remote Collaborative Mixed Reality.” M. Sajid, **S. Bukhari**, B. Ji, and B. David-John. *2025 IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR 2025)*.
- “Visceral Interfaces for Privacy Awareness of Eye Tracking in VR.” G. Ramirez-Saffy, P. Chelladurai, A. Vargas, **S. Bukhari**, E. Selinger, S. Foster, B. Heller, and B. David-John. *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2024)*.

ADDITIONAL EXPERIENCE

Graduate Teaching Assistant
Virginia Tech

January 2025 – Present
Blacksburg, VA

- Courses: System and Software Security, Professionalism in Computing, and Introduction to Programming.

Undergraduate Teaching Assistant
LUMS

September 2020 – August 2022

- Courses: Intro to Programming, AI, Software Engineering, Computer Vision Fundamentals, and Computational Problem Solving.

Full-Stack Software Engineer
3DevLab

March 2023 – August 2023

- Architected a cross-platform real-time contact-sharing application, authored design documents, and integrated decentralized components; launched on Google Play Store with active users.
- Advised 3+ clients on Web3-based data immutability while coordinating a cross-functional Agile team.

Backend Software Engineer
GoSaaS Labs

August 2022 – March 2023

- Developed seven cloud-to-network-drive features, improving data retrieval speed by 15%.
- Optimized SQL queries to reduce backend response time by 25%.

TECHNICAL SKILLS

HCI and Research Methods: Mixed methods, controlled experiments, participatory design, focus groups, scenario-based studies, usability testing, within-subjects designs, qualitative and quantitative data analysis, statistical analysis

Human-AI and Model Evaluation: Human-grounded evaluation, benchmark design, prompt robustness, AI alignment evaluation

XR and Prototyping: Unity, OpenXR, MRTK, HoloLens 2, Magic Leap 2, Meta Quest 3, Figma

AI and Data: Python, R, PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib

Software Engineering: C++, C#, SQL, JavaScript, Node.js, Express.js, Git, Docker, Microsoft Azure, Amazon Web Services

SERVICE & AWARDS

- **Mentoring:** Mentored Kim Giordano and Amulya Chinnala; UR2PhD-certified Graduate Student Mentor.
- **Reviewing:** *VRST'26*, and GenXR-ID workshop at *CVPR'26*.
- **Leadership & Service:** Virginia Tech Computer Science Graduate Council Webmaster (2024–25), Treasurer (2025–26), and Vice President (2026–27); iXR Ideathon organizing committee (2026).
- **Awards:** CCI Cyber Innovation Scholar (2 awards; \$3,500 total), PoPETs'26 stipend (\$840), and Dean's Honor List.